CORETRION™

32-Bit Hybrid Compute Processor

Product Brief

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Notice

This document provides a preliminary overview of the CORETRION™ processor architecture and product direction. Certain architectural capabilities, implementation details, and software support described herein are planned or under development and may evolve throughout product development. This document does not constitute a performance guarantee or final product specification.

Executive Summary

The increasing demand for intelligent embedded computing has created a need for processor architectures capable of efficiently supporting both traditional embedded workloads and emerging artificial intelligence applications. Conventional processor architectures often rely on external accelerators or heterogeneous compute devices to execute vector operations, machine learning inference, graphics workloads, and security functions, introducing additional system complexity and software overhead.

CORETRION™ is a proprietary **32-bit Hybrid Compute Processor** designed to address these evolving computational requirements through a modular architecture that integrates multiple specialized compute engines within a unified processing framework. Rather than targeting a single workload domain, CORETRION™ combines scalar processing with dedicated acceleration resources for vector computation, tensor operations, machine learning, graphics-oriented compute, and hardware-based cryptographic security.

The processor architecture is being developed with flexibility as a primary objective, enabling implementation across FPGA development platforms during prototyping while establishing a clear roadmap toward future ASIC realization. Its modular organization is intended to support incremental architectural evolution, allowing compute engines and system capabilities to expand over successive product generations without fundamentally altering the processor framework.

CORETRION™ is positioned as a versatile computing platform suitable for embedded intelligence, industrial automation, robotics, computer vision, digital signal processing, research, and FPGA-based system development. By integrating multiple computational domains into a cohesive architecture, the processor aims to simplify heterogeneous computing while providing a scalable foundation for future semiconductor innovation.

Technical Highlight

CORETRION™ integrates multiple specialized compute domains within a unified processor architecture, enabling a common execution framework for scalar computation, vector processing, tensor operations, machine learning acceleration, graphics-oriented compute, and hardware security.

Product Overview

Introduction

CORETRION™ is a proprietary processor architecture developed to support modern embedded and intelligent computing applications through an extensible hybrid compute model. The architecture combines a conventional scalar execution pipeline with dedicated accelerator subsystems that are designed to execute computationally intensive workloads more efficiently than general-purpose execution alone.

Unlike architectures that depend exclusively on external co-processors or software-managed accelerator interfaces, CORETRION™ is designed around an integrated execution philosophy in which specialized processing engines operate as coordinated components of the processor architecture.

The processor currently targets FPGA-based implementation for architectural validation, functional verification, software development, and ecosystem maturation. A future transition toward ASIC implementation is planned as architectural maturity increases.

Product Position

CORETRION™ is intended to serve as a flexible processing platform for applications requiring a combination of deterministic embedded processing and specialized computational acceleration.

The architecture has been conceived to address workloads involving:

- Embedded software execution
- Parallel vector computation
- Matrix-oriented operations
- Machine learning inference
- Graphics-oriented computation
- Secure embedded processing
- Real-time control systems
- Research and architectural exploration

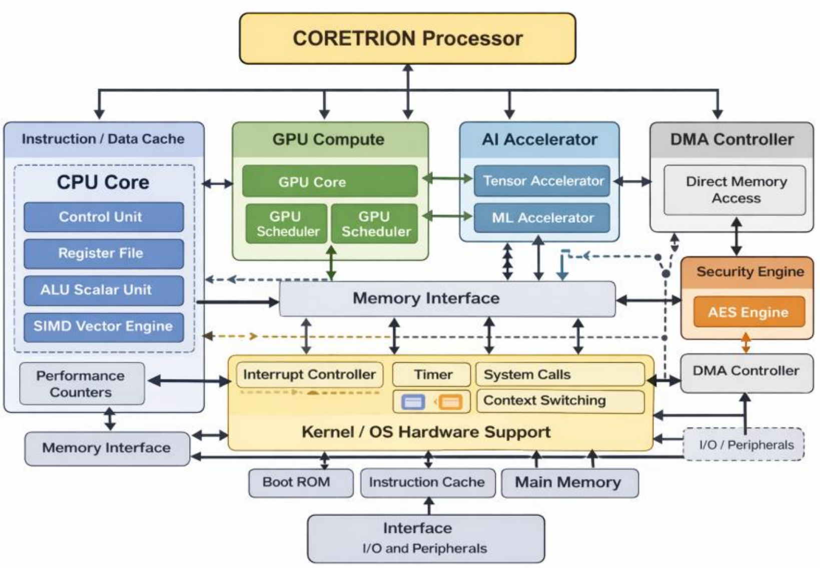
Rather than focusing exclusively on peak computational throughput, the design emphasizes architectural extensibility, modular integration, and efficient coordination between heterogeneous compute resources.

Product Objectives

The development of CORETRION™ is guided by several primary objectives:

Objective	**Description**
Unified Hybrid Compute	Integrate multiple compute engines within a single processor architecture.
Modular Expansion	Allow future architectural capabilities to be incorporated with minimal redesign.
FPGA Validation	Enable efficient prototyping, verification, and software development.
ASIC Readiness	Establish a structured path toward future silicon implementation.
Embedded Intelligence	Support emerging AI-enabled embedded applications.
Research Platform	Provide an extensible architecture suitable for academic and industrial research.

Figure 1. CORETRION™ Hybrid Compute Architecture



Design Principles

The architecture has been developed according to several foundational engineering principles.

Modularity

Each compute engine is designed as an independent architectural component with clearly defined interfaces, enabling future enhancement without requiring extensive redesign of unrelated subsystems.

Extensibility

Future processor generations are intended to incorporate additional compute capabilities, memory technologies, and interconnect mechanisms while preserving software compatibility wherever practical.

Balanced Computing

CORETRION™ seeks to balance scalar execution with specialized acceleration resources, enabling applications to utilize the most appropriate computational engine for a given workload.

Hardware Acceleration

Frequently encountered computational patterns—including vector arithmetic, matrix operations, and cryptographic processing—are intended to execute on dedicated hardware resources rather than relying exclusively on software implementations.

FPGA-Centric Development

FPGA implementation serves as the primary development platform for architectural verification, software enablement, hardware validation, and ecosystem development prior to future ASIC realization.

Technical Note

> CORETRION™ employs a modular hardware organization that enables progressive architectural evolution while maintaining a consistent execution framework across processor generations.

Vision and Design Philosophy

Engineering Vision

The evolution of embedded computing increasingly requires processors capable of executing diverse computational workloads within constrained power, area, and system complexity budgets. Applications in robotics, industrial automation, intelligent sensing, computer vision, and edge artificial intelligence frequently combine conventional software execution with computational kernels that benefit from specialized hardware acceleration.

CORETRION™ is being developed with the objective of creating a processor architecture that integrates these computational domains into a unified execution platform.

Rather than viewing artificial intelligence, vector processing, graphics computation, and security as isolated hardware components, the architecture considers them complementary elements of a coordinated compute ecosystem.

Design Philosophy

The CORETRION™ architecture is based on five primary design principles.

1. Unified Hybrid Computing

Traditional embedded processors typically execute all workloads through a general-purpose execution pipeline. While software flexibility remains valuable, computationally intensive algorithms often benefit from dedicated execution resources.

CORETRION™ incorporates specialized compute engines that are intended to operate as coordinated extensions of the processor architecture rather than as loosely connected peripherals.

2. Scalable Architecture

Scalability has been incorporated throughout the architecture to facilitate future product evolution.

Examples include:

- Additional accelerator engines
- Expanded memory hierarchy
- Enhanced interconnect technologies
- New instruction extensions
- Improved software capabilities

This modular approach supports future architectural refinement without fundamentally redefining the processor organization.

3. Software-Hardware Co-Design

Efficient processor architectures require close coordination between hardware capabilities and software tooling.

Accordingly, CORETRION™ is being developed alongside a software ecosystem that is planned to include:

- Compiler support
- Firmware components
- Runtime libraries
- Development tools
- Verification infrastructure
- FPGA development environment

Software enablement remains an important component of the overall product strategy.

4. Practical Innovation

Rather than introducing complexity solely for differentiation, architectural features are evaluated according to their practical applicability across embedded computing workloads.

Design decisions prioritize:

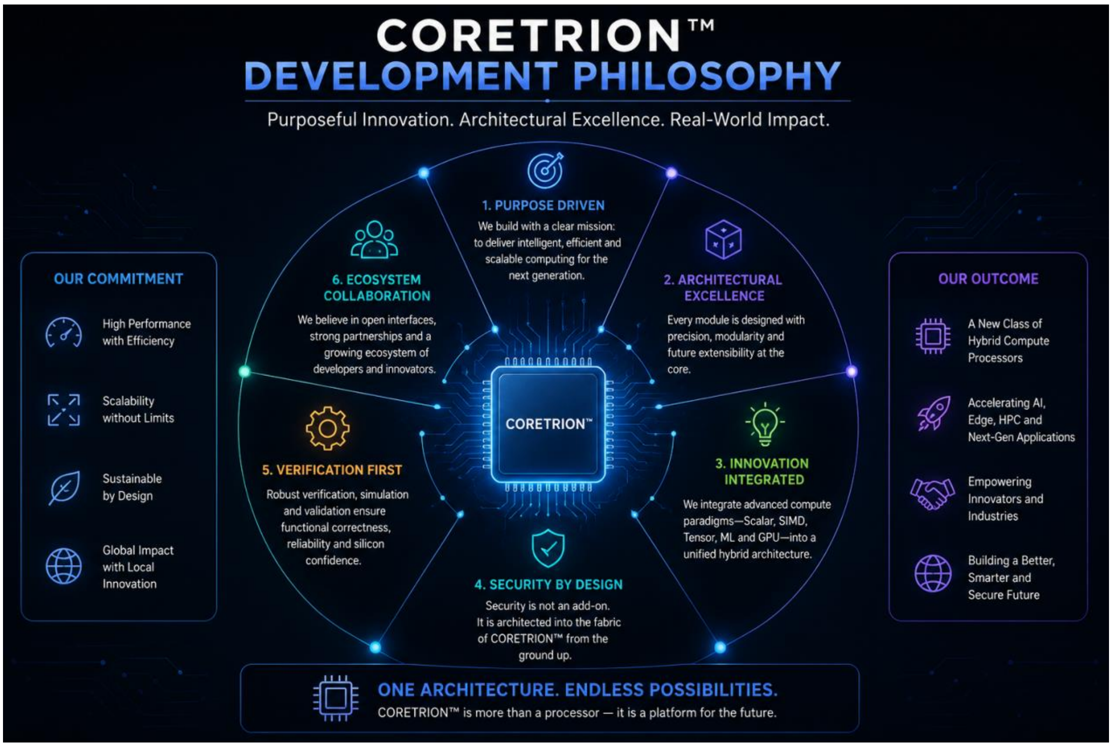
- Maintainability
- Verification
- Expandability
- Implementation feasibility
- Long-term architectural sustainability

5. Roadmap-Oriented Development

CORETRION™ is envisioned as the foundation of a broader semiconductor technology platform rather than a single standalone processor implementation.

Future development is planned to extend beyond the processor core itself toward complementary technologies including interconnect fabrics, chipsets, software tools, and development infrastructure.

Figure 2. CORETRION™ Development Philosophy



Architecture Principles

Principle	**Description**
Modular Design	Independent compute engines connected through a unified architecture.
Hybrid Computing	Scalar and accelerator execution integrated within a common processing framework.
Extensibility	Planned support for future architectural enhancements and compute capabilities.
Verification-Oriented	FPGA implementation enables functional validation and ecosystem development.
ASIC Transition	Architectural roadmap supports future silicon implementation.

Technical Highlight

CORETRION™ is not designed as a single-purpose processor. It is being developed as a modular hybrid compute platform capable of evolving across multiple generations of embedded and intelligent computing applications.

Why CORETRION™

Addressing the Next Generation of Embedded Computing

Modern embedded systems are undergoing a fundamental transformation. Devices that previously performed simple control and monitoring functions are increasingly required to execute complex computational workloads including artificial intelligence inference, real-time vision processing, sensor analytics, autonomous decision-making, and secure communication.

Traditional embedded processors remain highly effective for deterministic control tasks, but many emerging workloads require additional computational capabilities such as parallel processing, matrix operations, and specialized acceleration.

Current approaches commonly rely on combinations of:

- General-purpose processors
- External AI accelerators
- Dedicated graphics processors
- Separate security controllers
- Custom hardware blocks

While these solutions provide capability, they may introduce additional system complexity, increased integration effort, and software management overhead.

CORETRION™ is designed around a hybrid compute philosophy that combines multiple computational domains into a unified processor architecture.

Problems CORETRION™ Addresses

1. Increasing Compute Diversity

Modern intelligent devices execute a wide range of workloads:

- Control algorithms
- Mathematical computation
- Machine learning inference
- Image processing
- Signal analysis
- Cryptographic operations

A single execution model is often inefficient for all workload categories.

CORETRION™ addresses this challenge through a heterogeneous architecture that provides specialized engines optimized for different computational patterns.

2. Embedded AI Processing Requirements

Artificial intelligence is moving from cloud-based environments toward edge devices requiring:

- Lower latency
- Local decision-making
- Reduced communication dependency
- Improved privacy
- Real-time response

CORETRION™ incorporates dedicated processing resources intended to support AI-oriented workloads closer to the point of data generation.

3. System Complexity Reduction

Conventional heterogeneous systems often require multiple independent processors and communication interfaces.

A unified architecture can simplify:

- Hardware integration
- Software development
- System validation
- Platform optimization

CORETRION™ aims to provide a coordinated compute environment where multiple engines operate within a common processor framework.

4. Need for Flexible Development Platforms

Research institutions, startups, and industrial developers require adaptable computing platforms for:

- Architecture experimentation
- Algorithm development
- Hardware validation
- Prototype systems

The FPGA-oriented development approach of CORETRION™ enables early access to processor capabilities before future silicon implementations.

Target Workloads

CORETRION™ is designed for workloads where a combination of general-purpose processing and specialized acceleration provides architectural advantages.

Embedded Software Execution

The scalar processing unit provides a foundation for:

- Firmware execution
- Operating system support
- Control applications
- System management functions

Artificial Intelligence Inference

The integrated acceleration architecture is intended to support:

- Neural network inference
- Matrix operations
- Feature extraction
- Classification workloads

AI capabilities are under active development and optimized according to future software and hardware validation.

Computer Vision

Vision workloads frequently require:

- Parallel arithmetic
- Image transformations
- Filtering operations
- Feature processing

The SIMD, tensor, and GPU-oriented compute resources are designed to support these computational requirements.

Robotics and Autonomous Systems

Robotic platforms require coordination between:

- Real-time control
- Sensor processing
- AI decision systems
- Communication interfaces

CORETRION™ targets these mixed workloads through integrated heterogeneous processing.

Key Advantages

Integrated Hybrid Architecture

CORETRION™ combines multiple compute engines within a unified processor organization.

This enables applications to utilize:

- Scalar execution for sequential workloads
- SIMD execution for parallel arithmetic
- Tensor execution for matrix-based computation
- ML acceleration for AI workloads
- GPU compute for parallel processing
- AES hardware acceleration for security operations

Modular Compute Expansion

The architecture is designed around independent functional blocks connected through defined interfaces.

Benefits include:

- Easier verification
- Incremental enhancement
- Future accelerator integration
- Improved architectural flexibility

FPGA-to-ASIC Development Path

Architecture Design
↓
RTL Development
↓
Simulation & Verification
↓
FPGA Prototype
↓
Optimization
↓
ASIC Implementation Roadmap

Research and Innovation Platform

CORETRION™ can serve as a platform for:

- Processor architecture research
- AI accelerator experimentation
- Hardware/software co-design
- FPGA-based computing studies

Key Features

1. 32-Bit Proprietary Hybrid Architecture

Overview

CORETRION™ implements a proprietary 32-bit processor architecture designed around a hybrid execution model.

The architecture combines a conventional processor pipeline with specialized compute engines capable of accelerating specific classes of workloads.

The processor design includes:

- Dedicated instruction execution flow
- Accelerator-aware operation routing
- Modular compute engines
- Extensible instruction architecture

Architectural Concept

The processor separates workloads according to computational requirements.

Workload Type	Execution Resource
Sequential operations	Scalar Core
Vector arithmetic	SIMD Engine
Matrix computation	Tensor Engine
AI workloads	ML Engine
Parallel graphics computation	GPU Engine
Encryption/decryption	AES Engine

Technical Highlight

The hybrid architecture enables software workloads to dynamically utilize specialized hardware resources without requiring independent processing platforms.

2. Scalar Processing Core

Overview

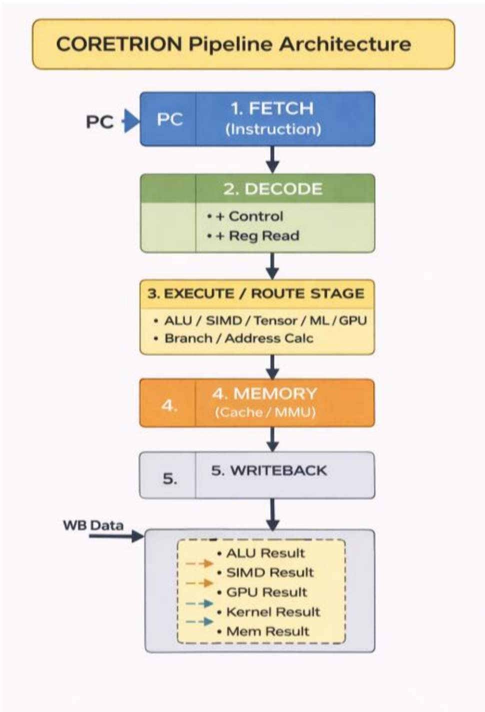
The Scalar Core provides the primary general-purpose execution capability of CORETRION™.

It is responsible for:

- Instruction execution
- Control flow management
- Integer operations
- System operations
- Memory coordination

Pipeline Organization

The scalar processing pipeline follows a structured execution model:



Responsibilities

The Scalar Core manages:

- Program sequencing
- Register operations
- Branch decisions
- Memory access requests
- Accelerator communication

3. SIMD Engine

Overview

The Single Instruction Multiple Data (SIMD) Engine provides parallel execution capability for operations where identical calculations must be performed across multiple data elements.

SIMD acceleration is intended for workloads including:

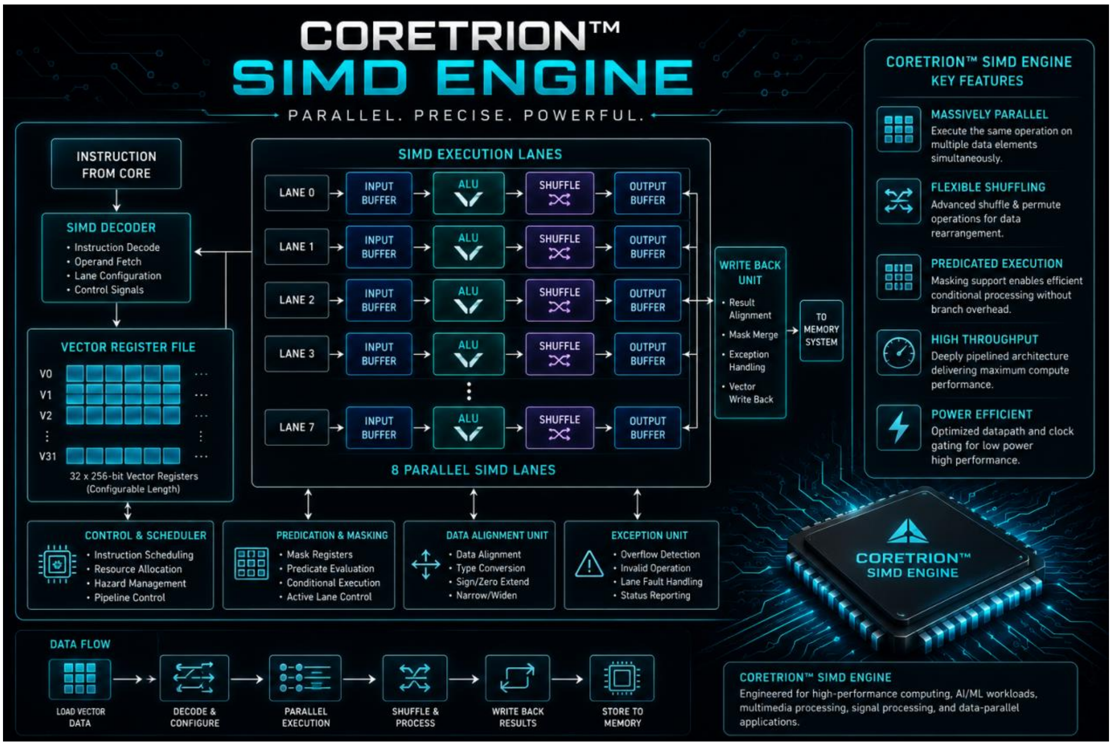
- Signal processing
- Image operations
- Vector mathematics
- Data transformation algorithms

Functional Advantages

Compared with sequential execution, SIMD processing enables:

- Increased data-level parallelism
- Reduced instruction overhead
- Efficient handling of repetitive mathematical operations

Figure 3. CORETRION™ SIMD Processing Model



4. Tensor Compute Engine

Overview

The Tensor Compute Engine is designed to accelerate matrix-oriented computations commonly found in artificial intelligence and machine learning workloads.

Tensor operations involve large numbers of multiply-accumulate operations, making them suitable for dedicated hardware acceleration.

Target Operations

The Tensor Engine is intended for:

- Matrix multiplication
- Neural network layers
- Convolution-oriented computation
- Data-parallel mathematical operations

Design Philosophy

The Tensor Engine is designed as a specialized compute resource rather than replacing the general-purpose processor.

The Scalar Core coordinates execution while the Tensor Engine performs computationally intensive operations.

5. Machine Learning Engine

Overview

The Machine Learning Engine provides dedicated support for AI-oriented workloads.

It is intended to enable:

- Neural inference acceleration
- AI model execution
- Embedded intelligence applications

Development Status

Machine learning capabilities are currently under development and will continue to evolve alongside:

- Hardware acceleration improvements
- Software libraries
- Model optimization techniques

6. GPU Compute Engine

Overview

The GPU Compute Engine provides parallel processing capability for workloads requiring large numbers of independent operations.

Potential applications include:

- Graphics computation
- Parallel algorithms
- Image processing
- Scientific workloads

Architecture Role

The GPU Engine operates as a specialized compute domain within the overall CORETRION™ architecture.

It complements:

- Scalar execution
- SIMD computation
- Tensor acceleration

by providing additional parallel processing capability.

7. AES Security Engine

Overview

Security is an essential requirement for connected embedded systems.

CORETRION™ incorporates a dedicated AES Security Engine designed to accelerate cryptographic operations.

Intended Functions

The AES Engine is intended to support:

- Hardware-assisted encryption
- Secure data processing
- Protected communication workloads

Conclusion

CORETRION™ represents a foundational approach toward developing a flexible and scalable hybrid compute processor architecture for next-generation embedded and intelligent computing applications.

By combining a general-purpose scalar processing core with specialized compute engines for SIMD processing, tensor computation, machine learning acceleration, parallel compute, and hardware security, CORETRION™ is designed to address the growing diversity of workloads encountered in modern embedded systems.

The architecture emphasizes modularity, extensibility, and hardware/software co-design, enabling future enhancements while maintaining a consistent processor framework. The FPGA-oriented development methodology provides a practical pathway for architecture validation, software enablement, and system-level experimentation, while establishing a structured roadmap toward future ASIC implementation.

CORETRION™ is being developed as more than a standalone processor core. It is envisioned as the foundation of a broader semiconductor technology ecosystem that includes future interconnect solutions, chipset platforms, software frameworks, and development tools.

As intelligent computing continues to expand across edge devices, industrial systems, robotics, and embedded platforms, CORETRION™ aims to provide an adaptable architectural foundation capable of evolving alongside future computational requirements.

Final Technical Statement

CORETRION™ is a modular 32-bit hybrid compute processor architecture designed to integrate scalar processing, specialized acceleration, and future semiconductor technologies into a unified computing platform. Through continued development, verification, and ecosystem expansion, CORETRION™ aims to establish a scalable foundation for future intelligent embedded computing systems.